

Chapter 2. PERMUTATIONS

2.1 What is a permutation?

2.1.3 Disjoint permutations



Today we will talk about **DISJOINT PERMUTATIONS**.

What are disjoint permutations?

We say that an element $k \in X_n$ is **fixed** by a permutation $\sigma \in S_n$ if $\sigma(k) = k$. Otherwise if $\sigma(k) \neq k$, we say that k is **moved** by σ . Two permutations σ and τ are **disjoint** if there is no element of X_n moved by both.

Examples 2.7.

1. The identity permutation $\varepsilon \in S_n$ is the only permutation that fixes all the elements of X_n .

2. $\begin{pmatrix} 1 & 2 & 3 & \dots & n-1 & n \\ 2 & 3 & 4 & \dots & n & 1 \end{pmatrix}$ moves every element of X_n .

3. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 5 & 4 & 1 \end{pmatrix} = \tau$ moves 1, 3, and 5, and fixes 2 and 4. τ and σ are disjoint permutations.

$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 3 & 2 & 5 \end{pmatrix}$ moves 2 and 4 and fixes 1, 3, 5.

Why are we interested in studying disjoint permutations?

THEOREM 2.8. *Disjoint permutations commute.*

LEMMA 2.9. If k is moved by σ , then $\sigma(k)$ is also moved by σ .

Proof

Let $k' = \sigma(k)$ and assume on the contrary that σ fixes k' . Then

$$\sigma(k') = k' = \sigma(k) \xRightarrow{\text{injective}} k = k' = \sigma(k)$$

$\Rightarrow k$ is fixed by σ , a contradiction. ■

Proof of Theorem 2.8. Let τ and σ be two disjoint permutations in S_n .

Goal $(\sigma\tau)(k) \stackrel{?}{=} (\tau\sigma)(k)$ for all $k \in \mathbb{Z}_n$.

Consider the following two subsets of $\mathbb{Z}_n$:

M_σ = set of elements of $\mathbb{Z}_n$ that are moved by σ

M_τ = set of elements of $\mathbb{Z}_n$ that are moved by τ

Since we are assuming that σ and τ are disjoint, we have $M_\sigma \cap M_\tau = \emptyset$.

For $k \in \Sigma_n$, we consider three cases:

Case 1 $k \in M_\sigma$

This means that k is moved by σ , and by the previous lemma we obtain that $\sigma(k)$ is also moved by σ , that is $\sigma(k) \in M_\sigma$.

Since $M_\sigma \cap M_\tau = \emptyset$ we have that $k \notin M_\tau$ and $\sigma(k) \notin M_\tau$. This implies $\tau(k) = k$ and $\tau(\sigma(k)) = \sigma(k)$.

$$\underline{(\tau\sigma)(k)} = \tau(\sigma(k)) = \underline{\sigma(k)}$$

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$$\Rightarrow \underline{(\tau\sigma)(k) = (\sigma\tau)(k)}$$

Case 2 . $k \in M_\tau$

Proceed like in Case 1.

Case 3 . $k \notin M_\sigma$ and $k \notin M_\tau$.

That means that $\sigma(k) = k$ and $\tau(k) = k$.

Thus:

$$(\tau\sigma)(k) = \tau(\sigma(k)) = \tau(k) = k$$

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